RetroPod – Project Documentation

A Retro-Inspired Spotify Music Player with Modern Functionality

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1. Introduction

RetroPod is a web-based music player that recreates the nostalgic experience of using a classic iPod while integrating modern music streaming through Spotify. The goal of the project is to combine the simplicity and iconic interaction style of early portable music players with the power of modern streaming platforms.

The application replicates the visual and functional aspects of a traditional iPod, including a digital display, menu-based navigation, and a click-wheel control system. Users can browse playlists, search for songs, and control playback through an interface inspired by the iPod devices of the early 2000s.

RetroPod is designed as a lightweight web application that users can keep open while working or browsing. By focusing on a minimal and nostalgic interface, the platform provides a more immersive and distraction-free way to enjoy music compared to conventional streaming interfaces.

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2. Project Vision

The main vision behind RetroPod is to revive the intuitive interaction design of classic music players while leveraging the extensive music library available through Spotify.

Modern streaming platforms often emphasize algorithm-driven feeds and complex layouts. RetroPod instead focuses on simplicity, allowing users to navigate music through a structured menu system similar to early digital music players.

The platform aims to deliver:

- Simple menu-driven navigation
- Retro hardware-inspired interaction
- Minimal and focused music playback experience
- Seamless access to modern streaming libraries

By blending nostalgic interface design with modern streaming technology, RetroPod provides a unique digital experience that celebrates the legacy of portable music devices while adapting them for modern web environments.

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3. System Architecture

RetroPod is built as a modern web application using a frontend framework integrated with Spotify services.

Frontend Interface

The frontend is responsible for rendering the iPod interface and handling user interaction.

Technologies used include:

- Next.js (React framework)
- TypeScript
- Tailwind CSS for styling
- Framer Motion for interface animations

The frontend manages:

- interface rendering
- menu navigation
- click-wheel interaction
- playback controls
- customization features

Spotify Integration

RetroPod integrates with Spotify through two main services.

Spotify Web API

This API retrieves music-related data such as:

- playlists

- albums
- artists
- tracks
- recently played songs
- music recommendations

Spotify Web Playback SDK

The Web Playback SDK enables music playback directly within the browser. It allows the application to control playback with features such as play, pause, skip tracks, volume control, and playback progress tracking.

Authentication

RetroPod uses Spotify's OAuth Authorization Code Flow with PKCE to securely authenticate users.

Authentication process:

1. User selects Connect Spotify
2. User logs into Spotify and grants access
3. Spotify redirects the user back to the application
4. The application receives an access token
5. Music playback and library access are enabled

This approach ensures secure authentication without requiring a dedicated backend server.

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4. User Interface Layout

The RetroPod interface is designed to closely resemble a physical iPod device. The layout is divided into two primary sections.

1. Display Screen

The screen acts as the main display area where menus, playlists, and playback information appear.

2. Click Wheel

The click wheel serves as the primary control system for navigating through menus and controlling music playback.

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5. Screen Interface and Navigation

Main Menu

The main menu functions as the primary navigation hub of the application.

Typical menu options include:

- Music
- Playlists
- Albums
- Artists
- Search
- Queue
- Now Playing
- Recommendations
- Settings

Each menu item opens a secondary interface containing related content.

Playlist and Track Navigation

Users can browse their Spotify playlists through the playlist menu. Each playlist displays its name, artwork, and number of tracks.

Selecting a playlist opens a track list showing:

- track title
- artist name

- track duration

The currently selected item is highlighted for easy navigation.

Now Playing Screen

The Now Playing screen displays detailed information about the current track, including:

- album artwork
- track title
- artist name
- playback progress bar
- elapsed time and duration

This screen allows users to monitor and control playback while navigating the interface.

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6. Click Wheel Interaction

The click wheel is the central interaction component of RetroPod and mimics the physical navigation system of classic iPods.

The wheel includes five functional regions:

- Menu button (top) – returns to the previous menu
- Back button (left) – plays the previous track
- Forward button (right) – skips to the next track
- Play/Pause button (bottom) – controls playback
- Center button – confirms a selection

Scroll Navigation

The circular scroll area detects rotational movement.

- Clockwise scrolling moves the menu selection downward
- Counterclockwise scrolling moves the selection upward

The active menu item is highlighted as the user scrolls through the list.

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7. Mini Player Mode

RetroPod includes a mini player feature that allows the interface to be displayed as a smaller floating player.

This mode allows users to control music playback while continuing to work in other applications or browser tabs.

Mini player features include:

- compact playback interface
- track information display
- play, pause, and skip controls
- draggable window positioning

The mini player can be activated using the Pop-Out Player option.

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8. Customization Features

RetroPod provides customization options that allow users to personalize their digital iPod.

Color Customization

Users can choose different color themes for the iPod body, such as:

- Classic White
- Space Grey
- Midnight Black
- Retro Pink
- Sky Blue
- Lime Green

Changing the color updates the device appearance while maintaining the retro design style.

Sticker Personalization

Users can decorate their iPod with digital stickers to create a personalized design.

Features include:

- selecting stickers from a predefined library
- dragging stickers onto the iPod body
- resizing and repositioning stickers
- saving the layout locally

This adds a playful customization element and enhances the overall interactive experience.

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9. Deployment

RetroPod is designed for easy deployment as a web application.

The deployment process typically includes:

1. Uploading the project to GitHub
2. Connecting the repository to Vercel
3. Configuring required environment variables
4. Deploying the application

Required environment variables include the Spotify Client ID and redirect URL used for authentication.

Once deployed, RetroPod can be accessed directly through a web browser.

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10. Conclusion

RetroPod demonstrates how classic hardware-inspired interaction designs can be reimagined using modern web technologies. By combining nostalgic interface elements with Spotify's powerful streaming capabilities, the application creates a unique and engaging music experience.

The platform highlights how minimal design and intuitive controls can coexist with modern digital services, offering users a focused and enjoyable way to interact with their music libraries.